



Question (1): (25 Marks)

A) [9 Marks] State whether each of the following statements is either True or False.

1. If you want to segment an image to the best number of segments, the best method to do segmentation using clustering would be K-means.
2. RANSAC is used to get rid of outliers in situations when there are wrong point correspondences.
3. The main parameter of the Mean-shift algorithm is the initial window position.
4. The number of Visual words is the main parameter of classifying images using BOW.
5. The task of image recognition is considered a sub-task of image verification.
6. Classifying an object incorrectly is considered a false positive error in object detection.
7. The YOLO model for object detection is more likely to produce multiple object responses in the image.
8. While applying down-sampling of an image in the hierarchical LK algorithm, the process of image sharpening would help avoid aliasing.
9. Normalizing the SIFT descriptors in the direction of the dominant gradient orientation would allow the descriptors to be invariant to image translation.

B) [16 Marks] Choose the correct answer

1. In Harris detector, the decomposition of the eigenvalues λ_1 and λ_2 of the second moment matrix M reveals the existence of a corner when:

- | | |
|--|------------------------------|
| a) λ_1 and λ_2 are small | b) $\lambda_1 \gg \lambda_2$ |
| c) λ_1 and λ_2 are large | d) $\lambda_2 \gg \lambda_1$ |

2. You are building a 3-class object classification and localization algorithm. The classes are: pedestrian ($c=1$), car ($c=2$), motorcycle ($c=3$). What would be the label for the following image? Recall $y=[pc, bx, by, bh, bw, c1, c2, c3]$



- | | |
|---|---|
| a) $y = [0, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$ | b) $y = [1, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$ |
| c) $y = [1, 0.3, 0.7, 0.5, 0.5, 1, 0, 0]$ | d) $y = [0, 0.2, 0.4, 0.5, 0.5, 0, 1, 0]$ |

3. Suppose two bounding boxes with sizes 2×1 and 2×3 are overlapping in a region of size 1×1 . The IoU between these two boxes will equal

- | | | | |
|----------|----------|-----------|-----------------|
| a) $1/6$ | b) $1/9$ | c) $1/10$ | d) none of them |
|----------|----------|-----------|-----------------|

4. Suppose you are using YOLO on a 19×19 grid, on a detection problem with 20 classes, and with 5 anchor boxes. During training, for each image you will need to construct an output volume y as the target value for the neural network; this corresponds to the last layer of the neural network. The dimension of this output volume will be

- | | |
|------------------------------|------------------------------|
| a) $19 \times 19 \times 45$ | b) $19 \times 19 \times 125$ |
| c) $19 \times 19 \times 500$ | d) none of them |

5. Suppose you run non-max suppression on the predicted boxes below. The parameters you use for non-max suppression are that boxes with probability ≤ 0.4 are discarded, and the IoU threshold for deciding if two boxes overlap is 0.5 . How many boxes will remain after non-max suppression?



$$\frac{2 \times 1}{2 \times 3} = \frac{2}{6}$$

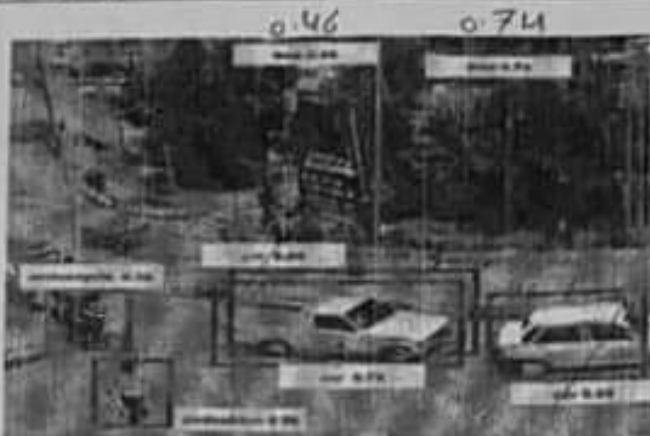


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6. The intuition behind detecting a corner in an image is that if the corner window slides for a small shift, the intensity of the window pixel values caused by the shift is
 a) low b) moderate c) high d) doesn't change
7. In a Siamese network, the definition of $\max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0)$ would be a correct definition for a triplet loss (where A represents the anchor image, P represents the positive image and N represents the negative image), considering that
 a) $\alpha = 0$ b) $\alpha < 0$ c) $\alpha > 0$ d) $\alpha \geq 0$
8. In the Bag of Words (BOW) algorithm used for local feature indexing in the context of object recognition from images, the number of k clusters selected by the k -means algorithm should be:
 a) representative to the number of visual words in the image dataset
 b) representative to the number of images in the dataset
 c) representative to the number of classes in the image dataset
 d) representative to the number features extracted from the image dataset
9. In an object detection model where it is supposed to produce a bounding box for each object that appears in the image, you consider a detection output as a false negative if
 a) $IoU > 0.5$ b) $IoU < 0.5$
 c) $IoU = 0.5$ d) no bounding box produced for the object
10. Using RANSAC for fitting a translational model, suppose we know that 30% of our data is outliers. How many times do we need to sample to assure with probability 80% that we have at least one sample being all inliers?
 a) 1 b) 1.3
 c) 2 d) 3

$$n > \frac{\log(1-p)}{\log(1-(1-\epsilon)^k)} \quad \epsilon = 0.3$$
11. A good feature descriptor should be:
 a) Repeatable b) distinctive
 c) Local d) All of the above
12. The Harris corner detector is stable under image
 a) Rotation b) translation
 c) Scale d) A and b
13. Instance segmentation is considered a hybrid technique between methods.
 a) Image classification and semantic segmentation b) image classification and object detection
 c) Sematic segmentation and object detection d) None of them
14. Visual Words help in encoding images by representing them as of feature occurrences.
 a) Ordering b) distribution
 c) Averaging d) random selection

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- B) [15 Marks]** Consider estimating the optical flow given two images $I(x, t)$ and $I(x, t+1)$. The brightness constancy constraint provides a linear equation for a given image pixel in the two consecutive images of an image sequence.
- What** is the brightness constancy constraint? **What** are its assumptions?
 - Explain** an algorithm for the optical flow estimation between the two images. **State** any extra assumptions you may need to let this algorithm work. **Show** the used equations in matrix form.
 - Given that the motion between the two images is non-linear, **what** are the approximated linearized version of your optical flow estimation algorithm that can be used to update the motion estimation? **State** the algorithm steps.



Fall 2023

Module Code: CN6141

Module Name: Robotics Vision and Learning

Course Code: AIT 417

Time allowed: 2:00 Hrs.

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Question (3): (25 Marks)

- A) [11 Marks] Sketch an architecture diagram of Faster R-CNN for object detection. Explain why it is called a two-stage object detector.
- B) [14 Marks] Sketch a general architecture of the Deconvolution network for semantic segmentation. Describe in detail the structure of the following: [6 marks]
- Input image. [2 marks]
 - Output image. [2 marks]
 - Final output layer of the network. [2 marks]
 - The network components and the function of each one. [2 marks]

Question (4): (25 Marks)

- A) [10 marks] Compare between Single Object Tracking (SOT) and Multiple Object Tracking (MOT) in terms of:
- Short/long term tracking.
 - Detection-free/Detection-based tracking.
 - Type of tracking errors.
 - Evaluations Metrics.
- B) [15 marks] One important application of autoencoders is Image denoising, where the Input Images are corrupted by adding some noise and feed them to the autoencoder to learn to remove the unwanted information. Sketch [2.5 marks] a general architecture of the denoising autoencoder showing the construction of the different layers of Encoder, Code and Decoder parts. Explain the role of each layer. [12.5 marks]

Examiners: Lecturer Dr. Dina Khattab

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